Talanta Open 3 (2021) 100026

Contents lists available at ScienceDirect

Talanta Open

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Electrochemical sensors and biosensors for the determination of diclofenac in pharmaceutical, biological and water samples

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ARTICLE INFO

Keywords:
Diclofenac
Electrochemical detection
Modified electrodes
Detection limit
Recovery

ABSTRACT

Diclofenac is a non-steroidal anti-inflammatory drug of wide use around the world for the treatment of several diseases such as ankylosing spondylitis, acute muscle pain conditions, and osteoarthritis. Although in proper doses it does not display any side effects, an overdose of diclofenac can cause adverse effects in the human body. Moreover, the presence of diclofenac in ecosystems directly affects the health of living organisms, even if it is present at trace. Therefore, the development of analytical tools for diclofenac monitoring was needed. In this context, electrochemical sensing and biosensing methods have been commonly reported. In this review, a critical evaluation of the methods employed for diclofenac analysis is presented according to the following classifications: potentiometric sensors, sensing at non-modified solid electrodes, chemically modified electrodes and biosensors. Recent developments related to the use of electrochemical detection of diclofenac in pharmaceutical formulations, biological fluids and environmental samples were reported and discussed.

1. Introduction

Diclofenac (DCF), designated as 2-(2-((2,6-dichlorophenyl) amino) phenyl) acetic acid, is one of the most frequently prescribed nonsteroidal anti-inflammatory drugs (NSAIDs) having antipyretic and analgesic properties [1]. Since 1974, DCF has been proved to be extremely efficient for the rheumatic complaints, acute joint inflammation and mild to moderate pain [2]. Because of its low solubility, DCF is often used in its sodium or potassium salt forms.

In general, DCF seems to be safe and appears to have no toxic effects on human's health when taken in normal therapeutic doses (50 to150 mg) [3]. However, taking high doses of DCF may cause adverse effects in the body such as aplastic anemia, gastrointestinal disorders and agranulocytosis and disturbs in renal function [4]. In addition, because of its wide use and poor biodegradation, DCF is considered as an emerging chemical contaminant that can cause adverse and unpredictable effects in the ecosystem [5]. It has also been reported to harm several environmental species at concentrations ≤1 μg/L [6]. Taking into account its consumption in pharmaceutical preparations and its negative impact on the environment and consequently on human health, the development of analytical procedures for DCF detection at low concentrations is required.

The number of papers published related to detection of diclofenac has steadily increased from 1967 to now. Indeed, the number of publications has been doubled during the last decade. Several conventional techniques were conducted to determine DCF, including high-performance liquid chromatography [7,8], gas chromatography [4], thin layer chromatography [9], capillary zone electrophoresis (CZE) [10], spectrophotometry [11,12], and spectrofluorimetry [13]. Generally, these methods are robust and precise but require expensive equipment and long analysis time. Alternatively, electrochemical methods are considered as a sample approach for DCF detection due to its high electroactivity [14–18]. These methods offer high sensitivity, specificity, low detection limit (LOD), low cost, user-friendly operation and simplicity. The poor response of bare electrodes in direct analysis of electroactive species has led to the use of mediators and modified electrodes to enhance the electrochemical response. Many research groups have developed innovative electrochemical sensors and biosensors for rapid and sensitive detection of DCF in various samples and matrices [15–19]. These sensors and biosensors are composed of an electrode as transduction element that covered with a chemical or biological recognition film.

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https://doi.org/10.1016/j.talo.2020.100026
Received 19 August 2020; Received in revised form 31 October 2020; Accepted 15 November 2020
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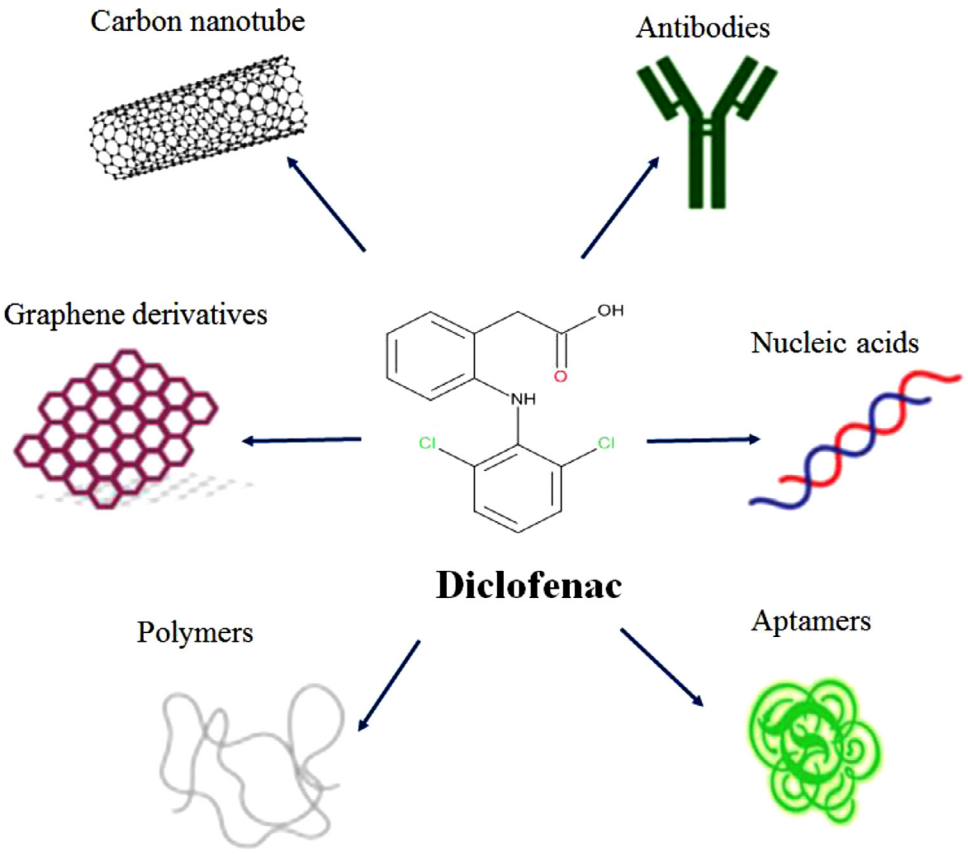


Fig. 1. Various sensing and biosensing materials used in the electrochemical detection of diclofenac.

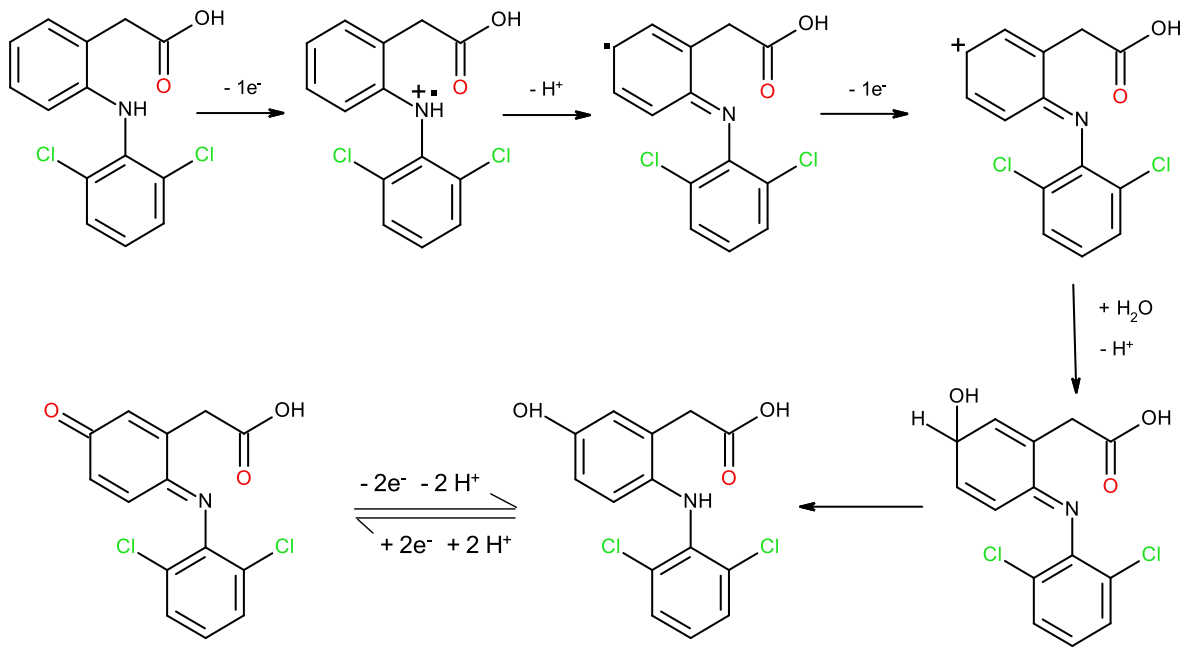
The specific recognition properties of this film led to the development of highly selective sensor.

In this paper, we comprehensively review the most promising mediators used for the sensing and biosensing of DCF as depicted in Fig. 1. The approaches used are evaluated based on the obtained performance parameters (linear range, limit of detection, precision of measurements, selectivity/interfering compounds), which are presented in several tables. The properties of the materials used for surface modification and electrocatalytic effects on the sensing and biosensing of DCF have been thoroughly discussed in this review. To the best of our knowledge, there is no review article related to the electrochemical determination of diclofenac.

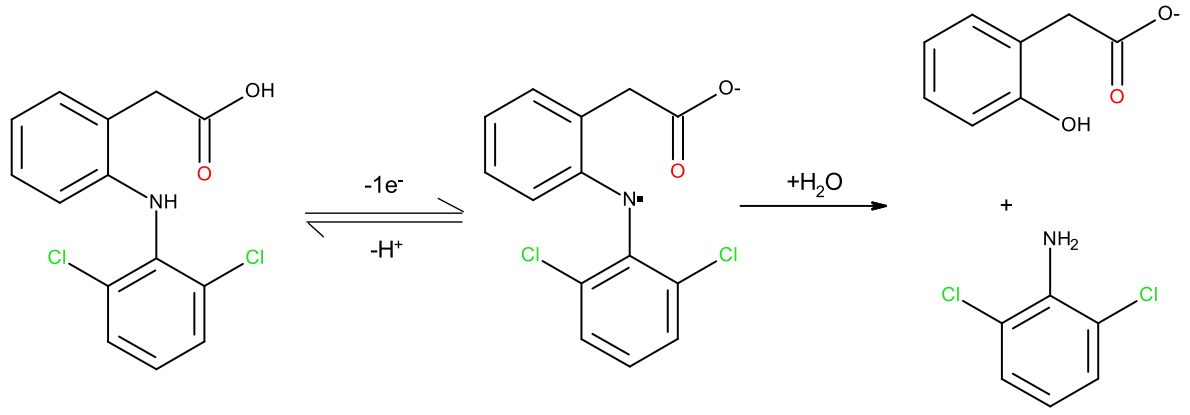
2. Diclofenac detection at non-modified electrodes

Many electrochemical investigations by unmodified electrodes have been reported for the analysis of diclofenac in pharmaceutical, biological and environmental samples [15,20–27]. Thanks to their low cost and low chemical reactivity, electrodes based on carbon materials still dominate in electrochemical analysis of DCF. The electrochemical oxidation of DCF on carbon-based electrodes offers the best analytical signals in the acidic or neutral medium. There exist very few studies about the mechanistic aspects of the redox processes of diclofenac. In 2010, *Goyal's* group [15] first proposed a possible mechanism of DCF oxidation at pH 7.2 at edge plane pyrolytic graphite electrode based on voltammetric data (Scheme 1). In that mechanism, DCF is irreversibly oxidized to 5-hydrohydiclofenac by a loss of $2e^-$, $2H^+$ process giving rise to an oxidation peak at ~662 mV when the sweep was initiated in the positive direction. 5-OH diclofenac is reduced in the successive cathodic sweep to diclofenac-2,5-quinone imine. During the second anodic sweep, a new oxidation peak is observed and related to the oxidation of diclofenac-2,5-quinone imine, involving $2e^-$ and $2H^+$. Later in 2016, *Cid-Cerón* et al. [28] studied the electrochemical oxidation of diclofenac at pH 7.0 using carbon paste electrode (CPE). The authors demonstrated that the electrochemical reaction is reversible and occurs through the loss of $1e^-$, with the formation of a radical intermediate and its subsequent hydrolysis to the final products 2,6 dichloroaniline and 2-(2-hydroxyphenyl) acetic acid, as shown in Scheme 2. *Aguilar-Liras* et al. [20] confirmed that diclofenac's oxidation at pH 7.0 using graphite electrode (GE) involves the formation of 2,6 dichloroaniline and 2-(2-hydroxyphenyl) acetic acid. This last presents a quasi-reversible redox equilibrium with 1-hydroxy-2-(hydroxyphenyl) ethanalate when the potential scan was inversed, and involves a one-electron transference, as shown in Scheme 3.

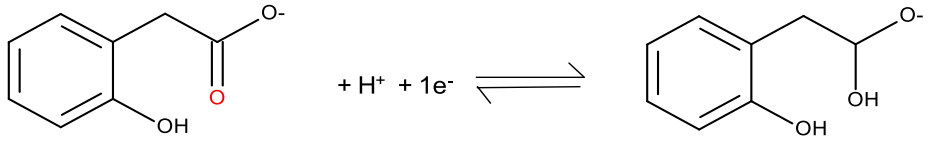
Bare carbon-based electrodes in combination with appropriate electrochemical techniques provide electroanalytical parameters fulfilling most of the analytical requirements. For example, a rapid and sensitive procedure for DCF analysis was developed on a boron-doped diamond electrode (BDDE) in 0.10 M sulfuric acid at pH 1.0 using flow injection method with multiple pulse amperometric detection [26]. The working concentration range of the procedure was 5 to 50 μM with a LOD as low as 0.14 μM. Also, this simple sensing platform was verified in pharmaceutical formulation with satisfactory recoveries from 98% to 102%. *Goyal* et al. [15] describes the DCF determination in phosphate buffer solution of pH 7.2 at the edge plane pyrolytic graphite electrode (EPPG) using square wave voltammetry (SWV). The electrode showed the linear range from 0.01 to 10 μM of DCF concentrations with a LOD of 6.2 nM. The applicability of the proposed procedure was also tested in pharmaceutical samples and urine samples obtained from patients undergoing treatment with diclofenac. The comparison of the analytical figure of merits of various unmodified electrodes for DCF analysis is provided in Table 1. As can be seen from the table, the diclofenac can be detected at submicromolar concentrations with unmodified carbon-based electrodes. However, bare electrodes exhibited high overpotential for DCF detection and slow electron transfer rate. The resultant traditional electrodes have low sensitivity and selectivity. At the same time, there is a



Scheme 1. A tentative mechanistic pathway proposed by Goyal et al. [15] for the electrooxidation of diclofenac at pH 7.2 at EPPGE.



Scheme 2. Oxidation mechanism for diclofenac anion on carbon paste electrode [28].



Scheme 3. Mechanism proposed for the electrochemical reduction of 2-(2hydroxyprop-2-enyl) acetic acid to 1-hydroxy-2-(hydroxyphenyl) ethanalate and its oxidation at pH 7 [20].

noticeable influence of the so-called ″chemical noises″ caused by impurities discharge currents and charging currents. In an attempt to resolve this problem, modification the surface of bare electrode was investigated in many research papers. This modification accelerates electrode kinetics and increases the sensitivity and selectivity of determination.

3. Electrochemical sensors for diclofenac determination

Modified electrodes have gained attention thanks to their high sensitivity and simplicity. An electrochemical sensor consists of an electrochemical transducer coated with a chemical film acting as conducting material [28]. Due to the high electroactivity of DCF, it can be detected at low concentrations using various types of modified electrodes based on advanced materials (Table 2). In fact, the use of theses electrodes allows enhancement of electrical signal and improvement of selectivity related to the oxidation of diclofenac.

3.1. Carbon nanotube-based electrochemical diclofenac sensors

Carbon nanotubes (CNTs) have been increasingly employed in many areas including fabrication of fast responsive sensitive sensors. Generally, two types of CNTs have been reported in literature namely multi-

Table 1
The comparison of analytical figure of merits of various unmodified electrodes used for DCF analysis.

Working electrode	pH	Technique	Linearity (μM)	LOD (μM)	Application/Recovery (%)	References
PPGE	7.2	SWV	0.01–10	0.0062	Tablet/ 96.64–98.95 Urine/96.67–102.13	[15]
GE	7.0	DPV	2.56–9.5	0.76	Tablets/-; Urine/-	[20]
GE	1	DPV	0.06–1	0.05	–	[21]
GCE	–	DPV	–	0.16	Tablets//98.0–102.0	[22]
GCE	–	DPV	0.5–6	–	–	[23]
BDDE	2.0	SWV	0.49–4.43	0.115	Tablet/97.25 Urine/103.0	[24]
BDDE	1.0	FIA system	5–50	0.14	Tablets/ 94.0–110.0	[25]
BDDE	H_2SO_4 (0.1 M)	BIA-MPA	10–50	1.1	Tablets/91.81;99.35	[26]
PPGE	8.2	CE-C^4D	250–1250	11	Tablets/90.08;98.92	[27]
	4.8	DPV	0.23–12.95	0.12	Tablets/ 95.20-101.34	

walled carbon nanotubes (MWCNT) [21,29–42] and single walled carbon nanotubes (SWCNT) [17,43]. Several types of electrodes have been modified with CNTs for the purpose of electrochemical DCF determination, as illustrated in Table 2.

Compared to the performance of other carbon electrodes, *Goyal* et al. [17] improved diclofenac sensitivity using an edge-plane pyrolytic graphite electrode (EPPGE) coated with single walled carbon nanotubes. Under the optimum conditions, the detection limit was 0.82 nM which was 10 times lower than that observed at bare pyrolytic graphite electrode. Furthermore, the electrode also provides high selectivity in the presence of an excess concentration of different interfering molecules and good recoveries in human blood plasma and urine samples. Functionalized CNT (fCNT) were also used for the construction of electrochemical sensors for the determination of DCF. For example, Slim et al. [31] modified a GCE with COOH-functionalized MWCNTs for amperometric detection of DCF at pH 7. The use of MWCNTs produced a significant improvement over the results obtained with bare GCE with a detection limit of 0.1 μM. The stability of the proposed sensor was examined for 10 days with acceptable RSD values. More recently, Sasal et al. [32], used screen-printed carbon electrode (SPCE) modified with carboxyl functionalized MWCNTs for DCF analysis. It was discovered that diclofenac was irreversibly oxidized at +555 mV when the sweep was initiated in the positive direction. In the second cycle, two reversible couples are observed consisting of two cathodic peaks and two anodic peaks. The modified electrode was applied for electrochemical detection of DCF in environmental water samples without sample pretreatment steps. Good results were obtained with a linear range of 0.1–10 nM and a limit of detection value of 28 pM. In addition to the overall good selectivity, no influence to DCF signal was observed in interference analyses. In another work reported by Yang et al. [34], a MWNTs–dihexadecyl hydrogen phosphate (DHP) film coated GCE was also used for the determination of diclofenac. It was discovered that the oxidation of diclofenac is a totally irreversible process with no peak appeared on the reverse sweep.

By combination of CNTs with other materials, the performance of the modified electrode can be significantly enhanced. Some studies have shown that the combination of CNTs with metal or metal oxide nanomaterials can increase electrochemically active sites and accelerate electron transfer kinetics. For example, Afkhami et al. [35] used MWCNTs and gold nanoparticles (AuNPs) for the determination of diclofenac sodium. The MWCNTs/AuNPs coated GCE (MWCNTs/AuNPs/GCE) was prepared by a rapid and simple electrodeposition method, as illustrated in Fig. 2i. MWCNTs were dispersed on the electrode without aggregation with special three-dimensional structures and smooth surface (Fig. 2ii (a)). The SEM image for AuNPs (Fig. 2ii (b)) shows a network-like structure of MWCNTs that AuNPs are located over the surface of it. In addition, the distribution of the nanoparticles was observed in the SEM image for MWCNTs/AuNPs/GCE exhibiting a large surface area. Differential pulse voltammograms of increasing DCF concentration for MWCNTs/AuNPs/GCE and the calibration curve has been shown in Fig. 2iii. Integration of Au nanoparticle to MWCNTs results in enhanced conductivity and electrocatalytic activity that presented remarkably good sensitivity with a detection limit of 0.02 μM. Moreover, the MWCNTs/AuNPs/GCE showed a stable and reproducible response without any influence of glucose, sucrose, fructose, citric acid and urea, but, acetaminophen interfered at 20 fold excess over DCF. The developed electrode was then successfully applied for the detection of DCF in human urine and diclofenac sodium tablet. Later, Akbarian et al. [43] reported a CPE modified with NiO functionalized SWCNTs and 2, 4-dimethyl-N/-[1-(2, 3- dihydroxy phenyl) methylidene] aniline as electro-catalyst for simultaneous determination of diclofenac, morphine and mefenamic acid. The mechanism of the oxidation of DCF at the proposed electrode is similar to that observed by Cid-Cerón et al. [28]. The modified electrode exhibits a linear response to DCF within the concentration range of 0.04–1200 μM, a low LOD of 8 pM, and good recovery rate when used for the detection of DCF in pharmaceutical samples without any interference.

Fig. 3.

CNTs were also applied with ionic liquids for surface modification to take advantage of synergistic effect. An electrochemical sensor based on MWCNTs and ionic liquid (HMIPF6) composite modified CPE was formulated by Ensafi et al. [36]. Under optimal conditions at pH 7.0, a detection limit of 0.2 μM was obtained. Another study reports the results of a CPE that has been modified with MWCNTs and ionic liquid (BMIMPF6) for the determination of diclofenac with a low LOD of 0.09 μM and high selectivity to cations, anions, and organic interfering influence [37]. Carbon ceramic electrodes (CCE) have also been modified with MWCNTs and BMIMPF6 for voltammetric determination of diclofenac at pH 7.0 [14]. The use of CNTs improves the analytical signal and enhances the effect of modification with the ionic liquid (BMIMPF6), resulting in a LOD of 18 nM, which is significantly lower than the LOD achieved in the previous examples. The oxidation reaction of diclofenac reported in all of these electrodes modified by ionic liquid and MWCNTs is similar to that observed by Goyal et al. [15].

Arvand et al. [38] reported a GCE modified with $Cu(OH)_2$ decorated MWCNTs for the electrochemical determination of DCF. In order to enhance the adhesion of MWCNTs/$Cu(OH)_2$ on the GCE surface, hydrophobic ionic liquid, namely 1-EMIMPF6, was used as a binder. Results demonstrate that the obtained electrode exhibits high catalytic activity toward the oxidation of DCF with a low LOD of 0.04 μM and good selectivity. Furthermore, this sensor demonstrates a potential practical application with good recovery rates when used for the detection of DCF in different real samples including seawater, fish serum, blood serum, tablet and ampoule. In another work, Damiri et al. [39] developed a voltammetric and impedimetric modified electrode for the selective detection of DCF in pharmaceutical formulations and urine samples based on an ionic liquid/cobalt hexacyanoferrate nanoparticle/MWCNTs modified GCE. The modified electrode exhibited excel-

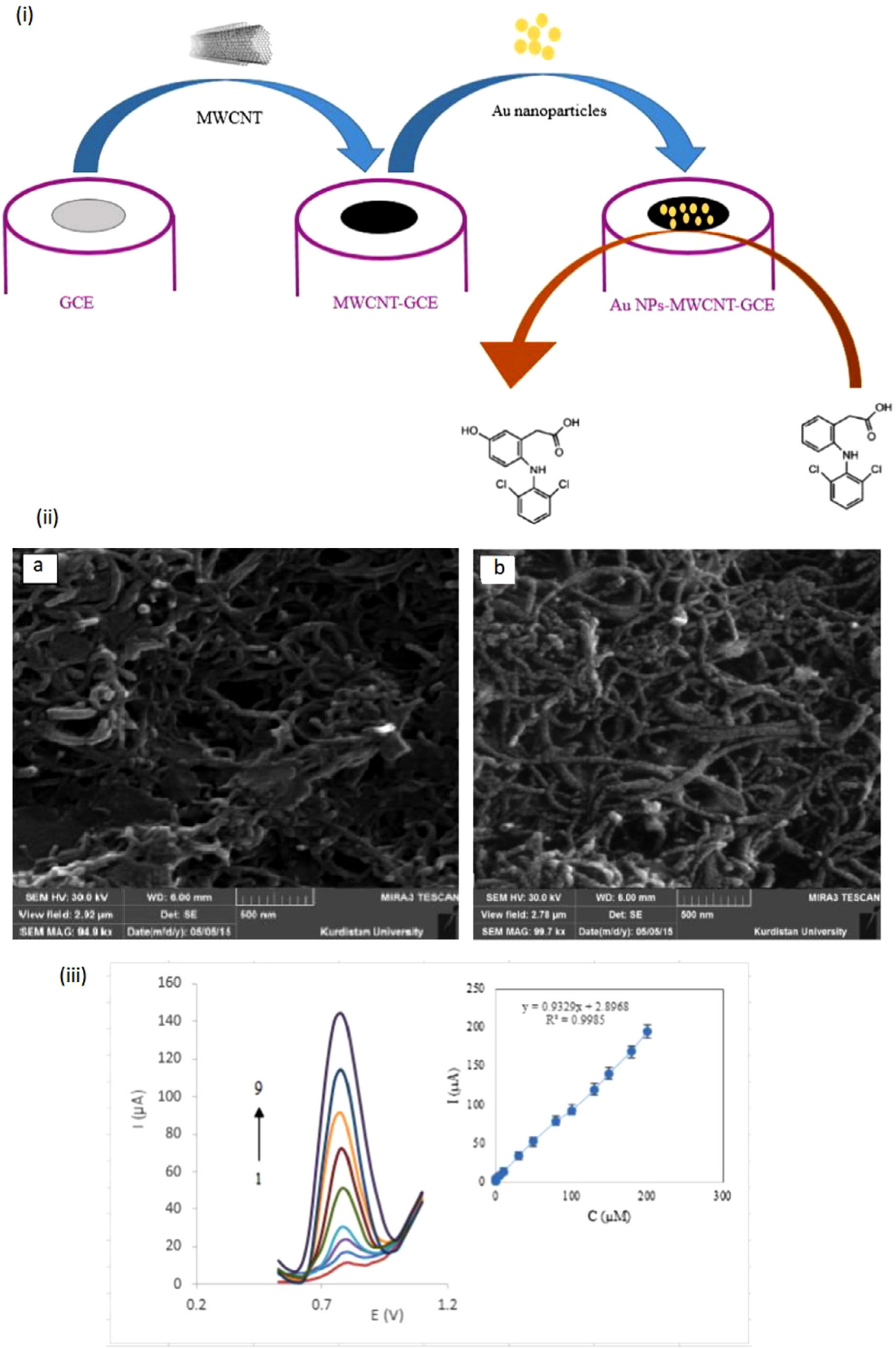


Fig. 2. (i) Schematic representation of sensor fabrication (ii) SEM images of modified GCEs: a) MWCNT/GCE; b) AuNP/MWCNT/GCE (iii) DPVs for different concentrations of DCF and calibration curve (inset). Reproduced with permission from Ref [35].

Table 2
Summary of the analytical performance data of the reported modified electrodes for diclofenac detection.

Working Electrode	Modifier	pH	Technique	Linearity (μM)	LOD (μM)	Application/ Recovery (%)	Ref
CCE	BMIMPF6- MWCNTs	7.0	DPV	0.05–50	0.018	Tablets/ 96–98.5 Plasma/ 97–99.6	[14]
EPPGE	SWCNTs	7.2	SWV	0.001–0.5 0.025–1.5	0.00082 0.0225	Tablets/96.12–98.1 Human blood plasma/54–118.33 Urine/90–132	[17]
CCE	BMIMPF6- MWCNTs	7.0	DPV	0.05–0.01	0.027	Human blood/ 97.9–101.6	[21]
CPE	MWCNT	8.0	DPV	0.1–90	0.74	Tablet/ 98.17	[29]
PGE	MWCNT	4.0	DPV	0.047–12.95	0.017	Tablets/ 95.80–104.28	[30]
GCE	COOH-MWCNTs	7.0	Amperometry	2–15	0.1	–	[31]
SPCE	COOH-MWCNTs	4.0	DPAdSV	0.0001–0.01	0.000028	River water/99.6–100.9	[32]
CPE	MWCNTs-vinylferrocene	7.0	SWV	5.0–600.0	2.0	Tablets and urine	[33]
GCE	MWNTs-DHP	5.0	LSV	0.17–2.5	0.08	Tablets/ 99.3–102.0	[34]
GCE	AuNP-MWCNT	4.0	SWV	0.03–200	0.02	Urine/ 97.5–100.5 Tablet/ 97.6	[35]
CPE	HMIPF6- MWCNTs	7.0	DPV	0.50–300	0.2	Injection solution/99.41–100.71 Tablets/98–104 Urine/99.46–100.6	[36]
CPE	BMIMPF6- MWCNTs	7.0	SWV	0.3–750	0.09	Injection solution/99.86;101.2 Tablets/97.5–106.0 Urine/99.5;102.0	[37]
GCE	MWCNTs/$Cu(OH)_2$/1-EMIMPF6	8.2	DPV	0.18–119	0.04	Ampoule/ 100.4 Tablet/ 100.7 Blood serum/100.0–100.5 Fish serum/99.1; 99.1 Seawater/98.3; 99.6	[38]
CPE	MWCNTs/CoHCF/BMIMCl	7.0	DPV	1.0–100.0	0.3	Tablet/ 96.1–102.0 Urine/ 98.2–101.6	[39]
GCE	MWCNT-CHT	4.0	SWV	0.3–200	0.021	Tablets/ 97.6–98.80 Urine/ 98.4–99.7	[40]
GCE	COOH-MWCNT-NC	4.0	DPV	0.05–250.0	0.02	Tablets/ 99.33–102.00 Ampoule/101.0–102.0 Urine/ 98.00–104.00 Serum/ 99.33–102.00	[41]
CPE	NiO-SWCNTs-DDPM	8.0	SWV	0.04–1200	0.008	Tablets/ 99.26; 102.86	[43]
GCE	COOH-GO	7.0	LSV	1.2–400	0.09	Urine/ 97.3; 99.2 Blood serum/ 96.7; 98.8 Tablets/ 96.8–99.0	[45]
SPCE	PT-RGO	7.0	DPV	0.1 – 100	0.04	Urine/ 84–105	[46]
CPE	$Co(OH)_2$-RGO	6.0	SWAdSV	0.025–1.55	0.008	Tablets/ 97.0–100.8 Ampoules/ 97.6–101.4 Serum/98.12–100.40 Urine/99.4–100.6	[47]
GCE	PDDA-Gr	7.0	DPV	10–100	0.609	Tablet/87.72 Lake water/95.40;102.55	[48]
GCE	Ag-poly-ACBK-Gr	2.5	DPV	0.1–40	0.034	Tablets/101.13 Capsule/101.84 Urine/99.60	[49]
GCE	poly(erichrome black T)	7.0	DPV	0.05–40	0.052	Tablet/ 96.37 Urine/ 97.82	[50]
GCE	4-Phosphatephenyl	7.4	DPV	0.62 - 1500	–	–	[51]
GCE	PEDOT-TiO_2	7.0	DPV	6–46	0.03	Tablets	[52]
CPE	PEDOT-TiO_2-[BMIM]Cl	6.0	DPV	5–100	0.011	Tablets/ 97.98–102.05	[53]
PGE	DCF-MIP	7.0	DPV	0.0054–0.54	0.0016	Urine/ 97.20–99.94 Urine/ 98.0–99.0	[54]
CPE	Polyanaline-RGO	–	DPV	16.88–27.35	3.71	Injection, tablet and human urine samples	[55]

lent electrocatalytic activity toward the oxidation of DCF. Under the optimized conditions, the linear response range and detection limit obtained by DPV method were found to be 1.0–100.0 and 0.3 μM, respectively.

In addition to metallic NPs and ionic liquids, polymers can also perfectly match with CNTs for the development of high-performance diclofenac sensors. Chitosan (CHT), a chemically inert biopolymer, is frequently used to mediate the immobilization of CNTs on carbon-based materials. Chitosan possessed hydrophilic and cationic behaviors, being suitable for the surface functionalization of CNTs with the formation of complex stable in water solution. It could wrap onto CNTs sidewalls through non-covalent interaction which help to improve the solubility of CNTs while preserving their intrinsic properties [44]. Shalauddin et al. [40] immobilized copper ions on MWCNTS-Chitosan thin film coated GCE (MWCNT-CHT-Cu/GCE). By using this modified electrode, the detection of diclofenac in human urine and pharmaceutical formulations was achieved with good results. In another work, nanocellulose (NC) as one of the most cost-effective, bio-degradable, biocompatible, environmentally friendly and easily available green nanomaterial was also used by Shalauddin et al. [41] in combination with f-MWCNTs to modify GCE. The electrode shows that the electrochemical oxidation of DCF is irreversible process with anodic peak at 0.677 V, which was lower than that of bare GCE. The prepared electrode showed a low LOD of 12 nM and was successfully tested in pharmaceutical and biological samples. In a summary, CNTs based modified electrode showed promising results for sensing of DCF. In all of the presented examples, DCF was

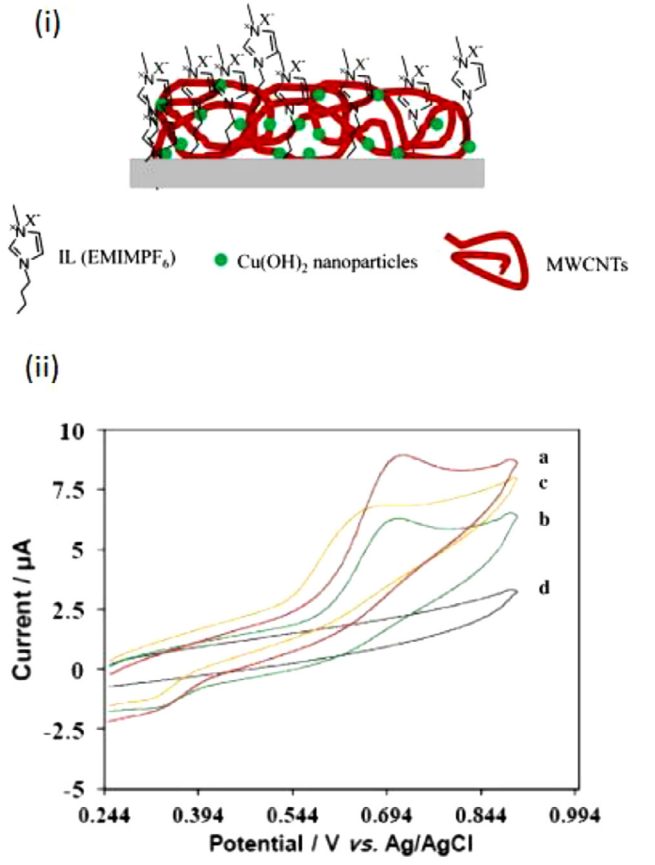


Fig. 3. (i) Schematic illustration of MWCNTs/$Cu(OH)_2$/IL modified electrode (ii) Cyclic voltammograms for MWCNTs/$Cu(OH)_2$ /IL nanocomposite (a), MWCNTs/$Cu(OH)_2$ (b) and MWCNTs (c) modified GCE in presence and absence (d) of diclofenac sodium. Reproduced with permission from Ref [38].

selectively detected from a complex matrix, including human serum, urine and commercial drugs. The best LOD results were obtained with SWCNT-modified edge-plane pyrolytic graphite electrode reproduced by Goyal et al. [17]. The main problem presented by these materials is the poor stability of CNTs suspensions in aqueous media, which reduce the probability of obtaining highly reproducible sensors. Considering the advantages and disadvantages of using CNTs as electrode modifiers, their functionalization creates more appealing features that assist their enhanced sensitivity, specificity and stable response.

3.2. Graphene and its derivative-based electrochemical DCF sensors

GR and its derivative-based composites have obtained extensive attention in the electrochemical detection of diclofenac (see Table 2). For example, Karuppiah et al. [45] fabricated an electrochemical sensor based on GO nanosheets functionalized with carboxyl group (COOH) for the detection of DCF. COOH groups on the surface of GO sheets exhibited an improved sensing response because this sensor has shown 1.6-fold higher anodic current than GO/GCE due to enriched carboxyl groups, superior mechanical strength, and electrical conductivity. Voltammetric data indicate slightly easier oxidation of DCF on such modified electrode (irreversible anodic peak at 580 mV vs. Ag/AgCl i.e. less positive compared to unmodified GCE). The modification of GCE electrode by COOH-GO resulted in greatly enhanced electrochemical oxidation current with a low detection limit of 0.09 μM. The device presented long term stability (retains 93.2% of current response after 30 days), satisfactory reproducibility and high selectivity. Furthermore, the electrode was applied to detect DCF in pharmaceutical and biological samples. Kimuam et al. [46] fabricated a modified electrode based on the composite of Pt and RGO, which was simultaneously synthesized via a single step electrochemical reduction on the screen-printed carbon electrode (SPCE). The electrode (Fig. 4i) demonstrated the well CV signal at 300 mV for DCF (Fig. 4ii). DPV results for different concentrations of DCF and calibration curve are shown in Fig. 4iii.The calibration plot was linearly proportional to the concentration of diclofenac over the range of 0.1 – 100 with a LOD of 0.04 µM. The Pt/RGO/SPCE showed a stable and reproducible response without any influence of glucose, citric acid, oxalic acid and ascorbic acid. However, the high concentration of histidine (≥10 fold) and dopamine (≥20 fold) significantly affected DCF sensitivity. The electrode was then successfully applied to detect DCF in human urine samples, with good recovery values in the range of 84–105%. El-Weki *et al.* [47] used CPE modified with RGO and $Co(OH)_2$ nano-flakes for simultaneous analysis of esomeprazole and diclofenac sodium in pharmaceutical and biological samples. The authors deduce the mechanism of diclofenac sodium reaction that agrees well with the mechanism reported by Goyal *et al.* [15]. Two electrochemical techniques, including square wave adsorptive stripping voltammetry (SWAdSV) and cyclic voltammetry (CV) were used to detect DCF with LOD of 8 nM and 50 nM, respectively.

Another approach for sensitivity enhancement is the incorporation of conductive polymers to fabricate nanocomposites of GR or its derivatives. Okoth et al. [48] constructed an electrochemical sensor based on the composite of poly(diallyldimethylammonium chloride) (PDDA) functionalized GR for the modification of GCE. The modified GCE was proposed for simultaneous determination of DCF and paracetamol at pH 7.0. The modified electrode demonstrated a better electrocatalytic activity towards DCF determination with satisfactory selectivity in the presence of paracetamol. The detection limit for DCF was found to be 0.609 µM and the sensor was used for the determination of DCF in pharmaceutical formulations and lake water. Recently, Huang et al. [49] fabricated a Gr-poly-acid chrome blue K functionalized Ag nanoparticles (Ag-poly-ACBK-Gr) nanocomposite on the surface of a GCE using layer-by-layer electropolymerization. A detection limit of 0.034 was obtained for DCF associated with a high selectivity in the presence of diverse interferents and good reproducibility, repeatability as well as high recovery values in drugs and human urine.

In summary, compared with Gr and GO, RGO has both conductivity and hydrophilicity with functional groups in its surface and it is easy to combine with other electroactive and conductive materials. The ability of RGO nanocomposite modified electrode was significantly improved in the application of DCF detection and the modified electrode had good selectivity, repeatability and stability.

3.3. Polymeric modifiers-based electrochemical DCF sensors

Conducting polymers have played a significant role in development of electroanalytical procedures. Hanabaratti et al. [50] fabricated a modified GCE surface by electropolymerization of poly(erichrome black T) using CV method. This electrode demonstrated that the electrochemical oxidation at pH 7.0 is an irreversible process with an anodic peak at 590 mV with no peak on the reversal sweep, which agrees well with the results reported by Yang et al. [34]. Using DPV method, the detection limit obtained was 0.052 µM. Exhaustive interference experiments revealed that the sensor possessed high level of selectivity. Another conducting polymer is 4-phosphatephenyl which has also been employed for the determination of diclofenac [51]. CV assisted electropolymerized 4-phosphatephenyl has been investigated for GCE surface modification. The oxidation peak of DCF is observed at about +650 mV, which corresponds to the oxidation of DCF to 5-hydrohydiclofenac. The calibration curve of the fabricated electrode exhibited a linear response from 0.62 to 1500 µM and was employed for its determination in a drug delivery system.

Combination of conducting polymers with advanced metal nanoparticles can synergistically improve the characteristics of sensing matrixes. Poly(3, 4-ethylenedioxythiophene) (PEDOT), used as an appro-

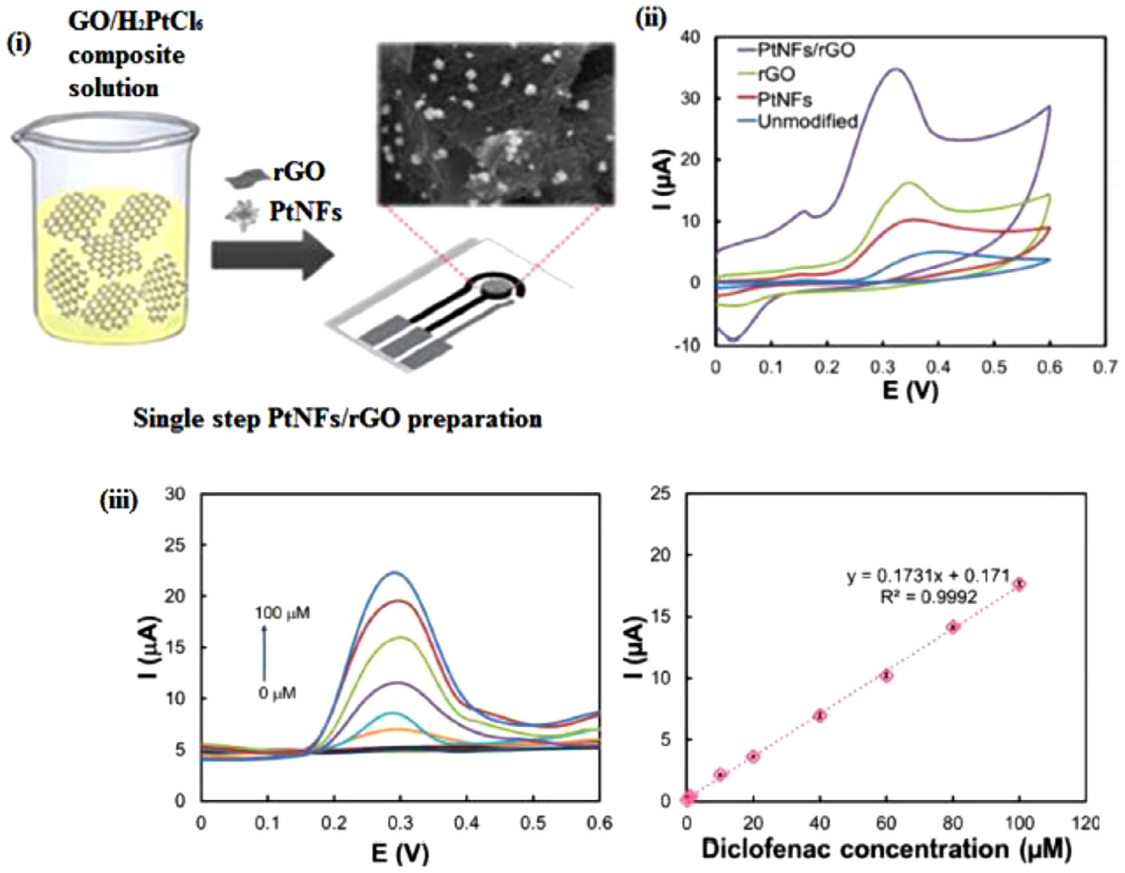


Fig. 4. (i) Schematic representation of electrode fabrication (ii) CV of 100 μM diclofenac measured on PtNFs/rGO, PtNFs, rGO and unmodified electrodes (iii) Differential pulse voltammograms of increasing DCF concentration, The inset shows the calibration curve of DCF. Reproduced with permission from Ref [46].

priate material for sensing, provided better results in combination with TiO_2 nanoparticles. Thiagarajan et al. [52] developed an electrochemical sensor based on electrodeposition of PEDOT on TiO_2 nanoparticles modified GCE surface for the simultaneous detection of diclofenac and ascorbic acid. The electrode was characterized with good electrocatalytic properties, excellent selectivity in the presence of ascorbic acid, long-term stability, excellent reproducibility, and good limit of detection as 0.03 μM. In order to enhance the adhesion of PEDOT/TiO_2 on the CPE surface, Farghali et al. [53] used ionic liquid, namely 1-butyl-3-methylimidazolium Chloride [BMIM]Cl. Results demonstrate that the obtained sensor exhibits high catalytic activity toward the electro-oxidation of DCF. The electrochemical oxidation of diclofenac at the proposed electrodes can be represented by a two-electron coupled two-proton transfer. Lower LOD of 0.011 μM for DCF detection was achieved at the developed sensor. Furthermore, this sensor was used to detect DCF in pharmaceutical and biological samples.

Conducting polymers when used individually as a modification material didn't show satisfactory results for DCF detection and the inferior long-term stability is another issue with these sensors. The conductivity and electrocatalytic properties of conducting polymers still have much room for improvement. These materials are used with nanomaterials to further improve the conducting, specificity and electrocatalytic properties of the modified electrode surface.

3.4. Molecularly imprinted polymers-based electrochemical sensors

Recently, the application of molecularly imprinted polymers (MIPs) based electrochemical sensors has been reported for the detection of various pharmaceutical compounds due to low cost of preparation, high specificity, good stability, and suitable sensitivity. The MIP electrodes are prepared via polymerization of certain functional monomers in the presence of template molecules (Fig. 5). This template is subsequently extracted from the resulting polymer matrix, generating a film filled with complementary cavities in the shape and size of the template molecule. For example, Fernandez-Llano et al. [54] reported a MIP modified electrochemical sensor for DCF determination. This sensing system was prepared by thermal polymerization over silica beads using 2-(dimethylamino)ethyl-methacrylate monomer. At pH 7.0, DCF is irreversibly oxidized at such sensor surface giving rise to an oxidation process at +540 mV, which was used as analytical signal. The MIP sensor exhibits a low limit of detection of 1.6 nM and good selectivity toward DCF. Furthermore, the results reveal that MIP sensor is appropriate for elimination of interferences from sample matrix. Another approach is the preparation of DCF-imprinted polymer for modification of CPE to fabricate a selective DCF electrochemical sensor as proposed by Mostafavi et al. [55]. The investigated MIP was synthesized by using polyaniline, a cross-linker (triphenylamine) and reduced graphene oxide. The fabricated sensor showed high recognition toward DCF compared to the non-imprinted polymer sensor. Though, the sensor is not very useful for DCF as detection limit is not satisfactory (3.71 μM).

4. Potentiometric sensors for diclofenac determination

Potentiometric detection based on ion-selective electrodes (ISEs) offer high precision and rapidity, low cost of analysis and wide range of concentrations. Moreover, they are easy to construct and manipulate and in most instances no pretreatment of the samples is needed before the analysis itself. Short response times of the order of seconds make ISEs appropriate devices for process control. Several approaches were reported for diclofenac using potentiometric sensors. A comparison of

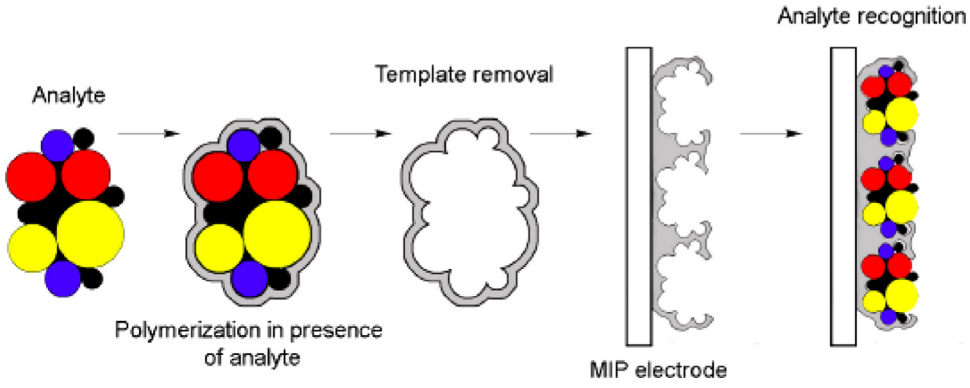


Fig. 5. Schematic representation of a molecularly imprinted polymer (MIP) electrode. Reproduced with permission from Ref [56].

Table 3
Figure of merits of the potentiometric sensors used in diclofenac analysis.

Ion pair	pH range	Linearity (μM)	LOD (μM)	Application/Recovery (%)	Refs
Ni(II) bathophenanthroline	8–12	50–1000	23	Tablets/ 98.1–100.4	[57]
2,4,6-Tri(2-pyridyl)-s-triazine Fe(II)	5.5–9.9	1–1000	1	Tablets/98.5–99.8	[58]
Fe(II) phthalocyanine	5.5–9	9–1000	9	Tablets/ 98.7–100	[59]
Fe(III) tetraphenyl porphyrin-chloride	6–9.9	10–1000	10	Tablets/99–100.49	[60]
Graphite matrix	6.5–9	50–5000	32	Tablets/96.8–101.0	[61]
Crystal violet	6–11	50–5000	25	Tablets/96.13–98.53	[62]
Safranine T	6–12	50–5000	32	Tablets/101.7–103.6	[63]
Basic dye BIK	7–11	100–500	100	Tablets/99.6–101.2	[64]
Neutral Red	5–12	50–5000	50	Tablets/92–102.6	[65]
Astrafloxacin FF	9–12	50–5000	50	Tablets/99.86–100.2	[66]
Mn(III)-tetrakis (3-hydroxyphenyl)porphyrin chloride	5.5–11.5	3–1000	3	Tablets/98.66–109.6	[67]
β-cyclodextrin	6.2–8.5	50–5000	14	Tablets/ 96.1–98.8 Urine/ 99.0–99.7	[68]
Imidazolium	–	10–1000	9.79	Spring water/97 Tablets/99–103	[69]
CV-Fe (β-CD)	7–10	01–1000	0.011	Tablets/ 98.38–101 Wastewater/-	[70]

analytical figure of merits for different research approaches for DCF detection by potentiometric sensors is provided in Table 3.

For the first time, in 1994, Hassan et al. [57] used a potentiometric sensor for the determination of diclofenac. The sensor is based on diclofenac-nickel(II) bathophenanthroline as an ion-exchanger site in a poly(viny1 chloride) (PVC) membrane plasticized with an intermediate o-nitrophenyl phenyl ether. At optimal conditions, the sensor exhibits a limit of detection of 23 µM and was used for the determination of DCF in pharmaceutical preparations. Others ISEs for DCF detection in pharmaceutical preparation and biological fluids were described either based on using cationic exchangers such as 2,4,6-tri(2-pyridyl)-s-triazine iron(II) [58], iron(II)-phthalocyanine [59], Fe(III) tetraphenylporphyrin-chloride [60], graphite matrix [61], crystal violet [62], safranine T [63], basic dye BIK [64], neutral dye [65], astrafloxin FF [66], manganese(III) tetrakis (3 hydroxyphenyl)porphyrin chloride [67], β-cyclodextrin [68].

Brennan et al. [69] constructed a potentiometric sensor based on imidazolium–diclofenac ion associate in a plasticized poly(vinylchloride) (PVC) membrane on planar carbon electrodes, with an intermediate of poly(3,4-ethylenedioxythiophene) layer. The sensor shows a detection limit of 9.79 µM and exhibits fast response, good stability, satisfactory reproducibility, and good selectivity for DCF and was applied successfully for the analysis of pharmaceutical gel and spiked spring water samples. More recently, another selective potentiometric sensor has been prepared by Elbalkiny et al. [70] based on functionalized nano-sized magnetized iron oxide (MNP-Fe_3O_4) particles using two cationic exchangers (Malachite green (MG) and Crystal violet (CV)) with β-Cyclodextrins (β-CD) forming sensors. The sensor based on CV exhibited a LOD of 0.11 µM, which is lower than those reported previously. Moreover, the sensor showed good selectivity toward DCF and was applied for the analysis of wastewater samples and pharmaceutical dosage forms.

5. Biosensors for the detection of diclofenac

Biosensors are modified electrode, which use biological materials as the recognition components. The biosensor is defined as a device that uses specific biochemical reactions mediated by isolated enzymes, immunosystems, tissues, organelles, microbes or whole cells to detect chemical compounds, usually by electrical, thermal or optical signals [71,72]. Electrochemical biosensors have been employed in several areas thanks to their high sensitivity (detection of low concentrations of analyte), simplicity (little equipment, few materials), potential for miniaturization and fast response time. Another important aspect for biosensors is selectivity, as it gives the particular affinity for observing specific target ions in the presence of other interfering ions. Selectivity of the biosensor is provided by the biorecognition component of the sensor which should be in high vicinity to transducing element. But, most importantly, they often do not need sample preparation and cleanup, which reduces time and cost of the analysis [73]. There are relatively

Table 4
Electrochemical biosensors for the detection of diclofenac.

Detection method	pH	Linearity (μM)	LOD (μM)	Application	Refs.
Amino-labeled aptamer/DCF/GCE	7.4	5–1000	0.27	Blood serum	[18]
NiHCF/PtNPs/acid-oxidized carbon nanotubes /polyethyleneimine	-	0.01–0.2	0.027	Tablets and human urine	[19]
DNA/GO/GCE	7.0	1–130	Not given	–	[75]
aptamer/Fe_3O_4/AuNP/CNT/GCE	7.4	1×10^{-8} to 1×10^{-6} 10^{-5} to 1.3×10^{-3}	3.40×10^{-9}	Tablets	[77]
ELISA (polyclonal diclofenac antiserum)	–	67.5–1350	20	tap and surface water	[80]
ELISA (monoclonal antibodies)	5.2	37×10^{-6} – 600×10^{-6}	26×10^{-6}	River, lake water and wastewater	[81]
Photon-upconverting NPs/antibodies	–	Not given	1.7×10^{-4}	Surface water samples and drinking water	[82]
DCF-Probe/E-Probe/GCE	7.8	1×10^{-8} - 1×10^{-4}	1×10^{-7}	Tap water	[83]

few studies addressing the development and the use of electrochemical biosensors for the determination of DCF (Table 4).

During the last decades, there was an emerging focus on the construction of DNA electrochemical biosensors. Deoxyribonucleic acid (DNA), biomacromolecule, has conductive properties which can be easily doped with electrocatalytic materials and has an enormous capacity to complex with a wide variety of small molecules [74]. The construction of DNA biosensor is based on the immobilization of nucleic acid on the surface of electrochemical transducer, where changes in DNA structure that occur during the intercalation of ligand molecules are detected, producing a measurable signal. For example, Wei et al. [75] had made an important contribution to the development of DNA biosensors for the detection of DCF. In such studies, GO was selected for the construction of a sensitive biosensor by immobilizing DNA on the electrode surface to further study the diclofenac-DNA interaction. The results show the irreversible oxidation of diclofenac with an anodic peak at 800 mV. On a DNA-GO modified GCE, the electrochemical signal was found to be linearly decreased with the increase of diclofenac concentration from 1 to 130 μM.

The electrodes modified with aptamer (artificial nucleic acids ligands) was also reported for the detection of DCF. Placement of the aptamer at the surface is usually mediated with the help of nanomaterials improving the electroanalytical performance of the sensors and sometimes facilitating the aptamer immobilization at the electrode surface [76]. For example, *Kashefi-Kheyrabadi and Mehrgardi.* [18] developed an amino-labeled aptamer, which is immobilized on the surface of a GCE. The whole procedure of biosensor construction was schematically presented in Fig. 6i. The immobilization of the aptamer on the surface of the surface of 6-aminohexanoic acid (AHA)/GCE and the loading of DCF on the amino-labeled aptamer (DBA) were monitored using electrochemical impedance spectroscopy (EIS) technique. The calibration curve of biosensor at pH 7.4 shows two different linear ranges over 0–5 μM and 10–1000 μM (Fig. 6ii). A detection limit of 0.27 μM and RSD mean of 4.7% was obtained. The aptasensor was then successfully investigated for the determination of DCF concentration in a human blood serum sample without interference from the sample matrix. In addition, the work investigates the changes in the plasmonic band of silver nanoparticles (Ag-NPs) upon the modification with the aptamer and DCF. The results are in line with the electrochemical experiments. Derikvand et al. developed a novel sensing nanocomposite with highly dispersed platinum nanoparticles (PtNPs) on carbon nanotubes (CNTs) functionalized with polyethyleneimine (PEI) as a platform for immobilization of diclofenac aptamer. Using electrochemical impedance spectroscopy (EIS), the proposed aptasensor showed a good detection range from 0.01 to 0.2 μM with an unprecedented detection limit of 0.027 μM and high selectivity [19]. In another study, *Azadbakht and Beirnvand* [77] used NH_2-functionalized Fe_3O_4/gold nanoparticles (Fe_3O_4/AuNP) decorated CNTs on the surface of GCE. Large amounts of amino groups in NH_2-functionalized Fe_3O_4 facilitate the immobilization of amine terminated DNA sequence (ssDNA1) by using glutaraldehyde as a linker. DCF aptamer as receptor against DCF (ssDNA2) was attached on the surface of sensing interface through the hybridization with ssDNA1. The recognition interaction between the aptamer strands and targeted molecule can cause decrease of the current observed by DPV using hexacyanoferrate as a redox probe. At pH 7.0, the aptasensor showed a low detection limit of 3.4 fM, linear ranges of concentration for DCF detection, from 0.01 to 1 pM and 10 to 1300 pM, and high selectivity and sensitivity. Moreover, the proposed method was applied for the determination of DCF in its pharmaceutical form with good recovery values in the range of 95.2–102.3%.

Immunosensors make another class of biosensors for the detection of a variety of analytes, which are fabricated using monoclonal (MAbs), polyclonal (PAbs) or recombinant (rAbs) antibodies as biorecognition elements [78]. Enzyme-based immunoassays (ELISA) techniques have been combined with biosensors, to form immunosensors, in order to increase their range, speed and sensitivity [79]. For example, Deng et al. [80] used an ELISA technique based on a polyclonal diclofenac antiserum prepared in rabbit for the determination of DCF at low concentrations with a limit of detection was 20 μM. The biosensor was applied for the determination of DCF in tap and surface water samples. Huebner et al. [81] development an ELISA procedure for the analytical determination of diclofenac based on monoclonal antibodies (mAbs). This aptasensor selectively detected DCF between (37–600) pM with a detection limit of 26 pM. The developed method was used for DCF residue quantification in water samples with satisfactory results. In another work, 0.17 nM in DCF concentration was reached by Hlaváček et al. [82], using antibodies according to a classical ELISA. The biosensor was also employed for the analysis of DCF in drinking and surface water samples. Another approach for DCF detection based on a novel biosensor was reported recently by Nguyen et al. [83]. This approach used an electrochemical displacement immunosensor based on the functionalization of a GCE by electroreduction of two functional diazonium salt, as shown in Fig. 7. The first one is a molecular probe (diclofenac) coupled with an arylamine in which a specific antibody is immobilized

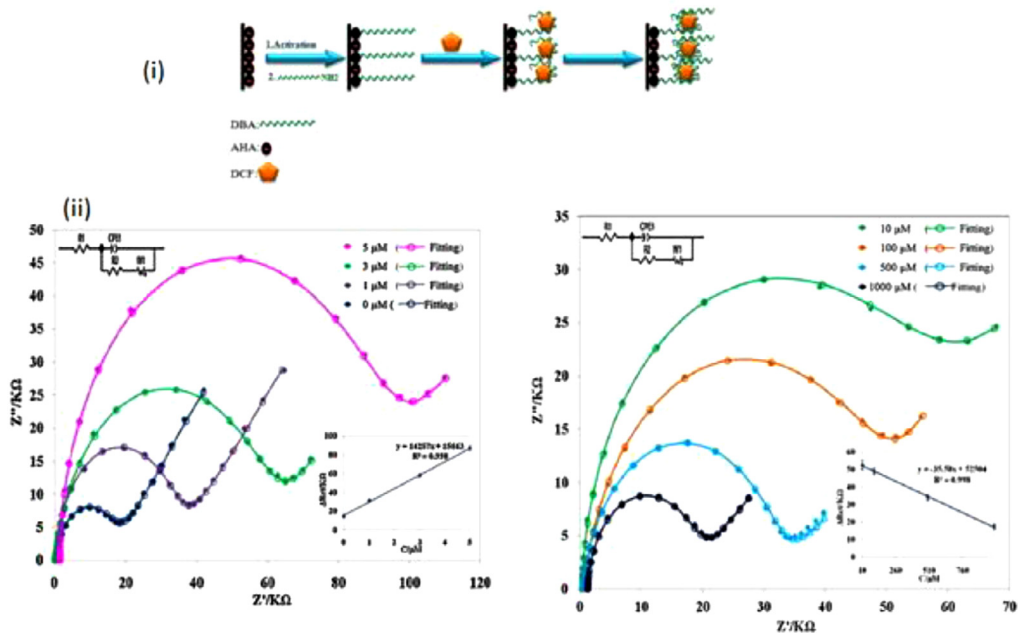


Fig. 6. (i) Schematic presentation of different steps of biosensor construction (ii) EIS complex plane plots for different concentrations of DCF and calibration curve (inset). Reproduced with permission from Ref [18].

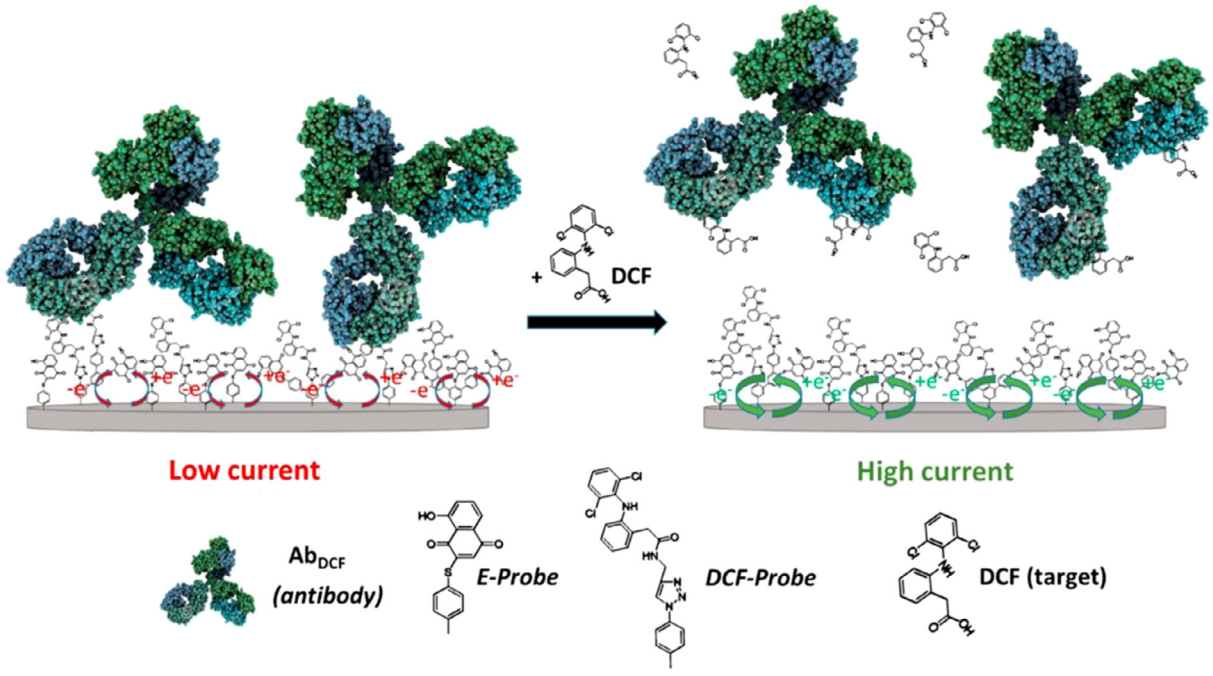


Fig. 7. Schematic illustration of the electrochemical displacement heterogeneous immunosensor. Reproduced with permission from Ref [83].

by affinity interactions; the second is a redox probe (a quinone) also coupled with an arylamine, able to transduce the hapten-antibody association into a change in electroactivity. The limit of detection was 0.1 pM, which is lower than that achieved in the previous examples. In a summary, it should be noticed that using nucleic acid aptamers present several advantages over antibodies, however, in the particular case of DCF detection, these aptasensor showed detection limits significantly higher than immunosensors.

6. Conclusion and perspectives

In this work, we have reviewed for the first time the several sensing approaches developed for the detection of DCF. Different modalities of sensor development were applied including potentiometric sensors, electrochemical sensors and biosensor. The development of potentiometric sensors is the earliest than the others, but their further research is limited by low sensitivity and high limit of detection values. Significant advances were reported for electrochemical determination of DCF using surface modified electrodes. The modification of electrode results in the large surface area and greatly electrocatalytical activity which enhance the analytical signal, diminishes the peak potential corresponding to DCF oxidation and reduced interference of other substances in actual samples. Modified carbon based-electrodes has been most extensively used for the detection of DCF among all working electrodes. In recent times, there has been a boom in the use of nanomaterials, including CNTs, GR and conducting polymer, as surface modifiers. The unique

and fascinating properties of these materials endow the electrochemical sensors with high sensitivity and selectivity, fast response rate, excellent stability and durability. Good results, in terms of sensitivity and LODs, are obtained with CNTs or GR, but the results obtained with conducting polymer-modified electrodes show even greater improvement. Achieving ultra-detection limit require the construction of biosensor using biomolecules specific for DCF, such as DNA or using aptamers or antibodies. Through there have been reported that the LOD of ultrasensitive DCF sensor can reach 0.1 pM [83], most of the sensors cannot reach this level.

Continuous monitoring of DCF in real samples will be advantageous compared to batch detection and efforts towards the development of continuous systems are encouraging. Most of the sensors were tested on real-life samples including pharmaceutical, biological fluids and environmental samples. Moreover, the reviewed examples are reflected in a variety of the proposed electrochemical approaches, based mainly on voltammetric (CV, DPV, and SWV) and to a lesser extent on amperometric (AMP and CAMP) techniques.

The review will be helpful to open new ideas in the field of (bio)sensors fabrication for more precise, sensitive, and selective detection of DCF by electrochemical techniques. Since the fabrication of sensors involve various currents of science, multidisciplinary effort will be helpful in developing ideal (bio)sensors. Although the progress in developing (bio)sensors is convincing, improvements in real sample analysis and selectivity still need to be focused on designing ideal sensing systems with better sensitivity and selectivity towards diclofenac. The integration of laser induced 3D bioprinting technology with microfluidic assisted electrochemical GSH sensing technology can improve electroanalytical performance compared to the current analytic systems, in terms of sensitivity, selectivity and detection limit. At this time no study reported the analysis of diclofenac by this technique.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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